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$$\int \frac{1-5x}{1-2x} dx$$

We herschrijven de teller:

$$1-5x = -(5x-1), \quad 1-2x = -(2x-1)$$

Dus:

$$\int \frac{1-5x}{1-2x} dx = \int \frac{-(5x-1)}{-(2x-1)} dx = \int \frac{5x-1}{2x-1} dx$$

$$\text{Stel } u = 2x-1 \Rightarrow du = 2dx \Rightarrow dx = \frac{1}{2} du$$

Substitueer $x = \frac{u+1}{2}$ in de teller $5x-1$:

$$5x-1 = 5 \cdot \frac{u+1}{2} - 1 = \frac{5u+5-2}{2} = \frac{5u+3}{2}$$

Dus de integraal wordt:

$$\int \frac{5x-1}{2x-1} dx = \frac{1}{2} \int \frac{5u+3}{u} du = \frac{1}{4} \int \frac{5u+3}{u} du$$

Splits de breuk:

$$\begin{aligned} &= \frac{1}{4} \left(\int 5 du + \int \frac{3}{u} du \right) \\ &= \frac{1}{4} (5u + 3 \ln |u|) + C \end{aligned}$$

Substitueer $u = 2x-1$ terug:

$$\begin{aligned} &= \frac{1}{4} (5(2x-1) + 3 \ln |2x-1|) + C \\ &= \frac{5}{2}x - \frac{5}{4} + \frac{3}{4} \ln |2x-1| + C \end{aligned}$$

References